

# Multiply de re counterpart theory and actuality

Handout for *Time and Modality* seminar

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## 1 Counterpart theory and arguments for coincidence

- (1) a. Lump1 could have survived squashing.  
b. Goliath could not have survived squashing.  
c. Therefore, Goliath is not Lump1.
- (2) a. No statue in the room has ever been ball-shaped  
b. Some statue-shaped lumps of clay in the room have been ball-shaped  
c. Therefore, some statue-shaped lumps of clay in the room are not statues

Distinguish two strategies for resisting such arguments:

- (i) Contextualist strategy: the arguments preserve truth-in-a-context, but there is a tendency for the context to shift from one premise to another.
- (ii) Logically revisionary strategy: the arguments are simply invalid. (It's not just Leibniz's Law that goes - syllogistic too!)

## 2 Counterpart theory and the necessity of identity

A putative counterexample:

- (3) Goliath is identical to Lump1, but it could have been the case that Goliath was distinct from Lump1.
- (4) Lump1 could have survived squashing, whereas Goliath could not have survived squashing.

## 3 The necessity of distinctness

Williamson's argument:

$$\begin{aligned}x \neq y &\rightarrow \Box \neg \text{ACT } x = y \\ \Box \text{ACT } x &= x \\ \Box ((\text{ACT } x &= x \wedge \neg \text{ACT } x = y) \rightarrow x \neq y) \\ \text{Hence: } x \neq y &\rightarrow \Box x \neq y\end{aligned}$$

## 4 Recap: counterpartings

COUNTERPAIRINGS

$$\begin{aligned}\Diamond \phi(t_1, \dots, t_n) &:= \exists R \exists v_1 \dots \exists v_n (\text{CR} \wedge R t_1 v_1 \wedge \dots \wedge R t_n v_n \wedge \phi(v_1, \dots, v_n)) \\ \Box \phi(t_1, \dots, t_n) &:= \forall R \forall v_1 \dots \forall v_n ((\text{CR} \wedge R t_1 v_1 \wedge \dots \wedge R t_n v_n) \rightarrow \phi(v_1, \dots, v_n))\end{aligned}$$

Require counterpairings to be *functional*  $((CR \wedge Rxy \wedge Rxz) \rightarrow y = z)$  and to have *universal domains*  $(CR \rightarrow \forall x \exists y Rxy)$ . So we can rewrite more compactly:

$$\begin{aligned} \text{COUNTERPAIRINGS}^* \quad & \Diamond \phi(t_1, \dots, t_n) := \exists R (CR \wedge \phi(R[t_1], \dots, R[t_n])) \\ & \Box \phi(t_1, \dots, t_n) := \forall R (CR \rightarrow \phi(R[t_1], \dots, R[t_n])) \end{aligned}$$

## 5 Counterpart theory and propositional contingentism/temporalism

*Truth Schema:* The proposition that  $\phi$  is such that necessarily, it is true iff  $\phi$ .

How to get the Truth Schema to come out valid in a counterpart theory?

*Proposition Constraint:* If  $R$  is a counterpairing, and  $R\alpha_1 x_1 \dots$  and  $R\alpha_n x_n$ , and  $p$  is the proposition that  $\phi(\alpha_1 \dots \alpha_n)$ , then  $Rpq$  is the proposition that  $\phi(x_1, \dots, x_n)$ .

## 6 Counterpart theory and elitism

Intuitive hallmarks of the modal and temporal elitists that don't mention worlds and times:

- The class that *actually* contains all and only the  $F$  objects is more natural than almost any other class that *could* have contained all and only the  $F$  objects.
- The class that *now* contains all and only the  $F$  objects is more natural than almost any other class that *sometimes* contains all and only the  $F$  objects.

Counterpart theorists should accept lots of claims like this!

Still, to properly address the question of modal and temporal elitism as I have formulated them, we had better ask what plays the role of worlds and instants in this theory.

One suggestion that I spent a lot of time thinking about: *worlds/instants are counterpairings*, and 'at' is understood as follows:

$$(\text{At } R) \quad \text{at } R \phi(t_1, \dots, t_n) := \phi(R[t_1], \dots, R[t_n])$$

It's certainly true that counterpairings play *many* of the canonical roles of worlds/instants, and that for many of the purposes where philosophers are used to quantification over worlds/instants, quantification over counterpairings would do just as well. (*Example:* the analysis of counterfactuals in terms of closeness.) However there are several problems with the proposal, some deeper than others.

- (i) In the way I've set things up, counterpairings aren't officially *entities* at all—quantification over counterpairings is plural quantification over ordered pairs.
- (ii) There's nothing I've said that rules out the possibility that there are some counterpairings  $R$  and  $S$  such that  $R \circ S^{-1}$  is a qualitative symmetry (preserves all qualitative properties and relations). But in that case, exactly the same propositions will count as being true 'at'  $R$  and  $S$ . If we want each instant to be present at itself and no other instant—or each world to be actualised at itself and no other world—we can't have that.

- (iii) Assuming that ordered pairs are rigid, counterpairings don't satisfy the crucial principles of (Permanence)/(Invariance):  
 (Permanence) If at  $t$ ,  $\phi$  then it is always the case that at  $t$ ,  $\phi$   
 (Invariance) If at  $w$ ,  $\phi$  then it is necessary that at  $w$ ,  $\phi$

A better idea: worlds/instants are just certain strong propositions. 'At  $w$ ,  $\phi$ ' means 'It is metaphysically necessary that if  $w$  is true then  $\phi$ '. 'At  $t$ ,  $\phi$ ' means 'It is always the case that if  $t$  is true,  $\phi$ '.

- In the modal case, the propositions need to be strong enough to metaphysically necessitate everything else or its negation. If we think that any propositions (plural) have a conjunction, we can say that a world is a proposition that is the conjunction of some propositions such that every proposition that is not one of them is such that its negation is one of them. However this is controversial.
- In the temporal case, we don't want the propositions to be *that* strong, since we want it to be possible that  $t$  is sometimes accurate even though the course of history is different. How is this going to work? It may be quite vague which propositions are the instants.

This vindicates modal/temporal elitism assuming that *truth* is a natural property.

A worry: we would like a theory which allowed us to *prove* instances of the schema

If  $\phi$ , then the actualised world  $w$  is such that (metaphysically) necessarily, at  $w$ ,  $\phi$

Consider the case where  $\phi$  is just ' $a$  is  $F$ '. Given COUNTERPAIRINGS and the proposed modal definition of 'at  $w$ ' this becomes

If  $a$  is  $F$ , then the true world  $w$  is such that for any counterpairings  $R$  and  $S$ , if  $R[S[w]]$  is true, then  $R[S[a]]$  is  $F$ .

We can't read this off straightforwardly from *Proposition Constraint*. *Proposition Constraint* only tells us about propositions that are only "about" *finitely many* objects. But on the proposed account, worlds don't seem to be like that—they are about infinitely many objects.

## 7 Is this really the A-theory?

What could be missing?

- (i) TRUE DYNAMIC CHANGE!!
- (ii) Fundamental tense operators?
- (iii) Temporary being?
- (iv) Open futurism?

## 8 What is special about metaphysical necessity, for the modal counterpart theorist?

(discuss if there is time)